Chapter 15

1. This statement ignores hidden-variable approaches, such as Bohm's. But even in such approaches, we'd want to teleport an object's quantum state (its wavefunction), so a mere measurement of position or velocity would be inadequate.

2. Zeilinger's research group also included Dick Bouwmeester, Jian-Wi Pan, Klaus Mattle, Manfred Eibl, and Harald Weinfurter, and De Martini's has included S. Giacomini, G. Milani, F. Sciarrino, and E. Lombardi.

3. For the reader who has some familiarity with the formalism of quantum mechanics, here are the essential steps in quantum teleportation. Imagine that the initial state of a photon I have in New York is given by $|\Psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1$ where $|0\rangle$ and $|1\rangle$ are the two photon polarization states, and we allow for definite, normalized, but arbitrary values of the coefficients. My goal is to give Nicholas enough information so that he can produce a photon in London in exactly the same quantum state. To do so, Nicholas and I first acquire a pair of entangled photons in the state, say $|\Psi\rangle_{23} = (1/\sqrt{2})\,|0_2 0_3\rangle - (1/\sqrt{2})|1_2 1_3\rangle$. The initial state of the three-photon system is thus $|\Psi\rangle_{123} = (\alpha/\sqrt{2})\,\{|0_1 0_2 0_3\rangle - |0_1 1_2 1_3\rangle\} + (\beta/\sqrt{2})\,\{|1_1 0_2 0_3\rangle - |1_1 1_2 1_3\rangle\}$. When I perform a Bell-state measurement on Photons 1 and 2, I project this part of the system onto one of four states: $|\Phi\rangle_{\pm} = (1/\sqrt{2})\,\{|0_1 0_2\rangle \pm |1_1 1_2\rangle\}$ and $|\Omega\rangle_{\pm} = (1/\sqrt{2})\,\{|0_1 1_2\rangle \pm |1_1 0_2\rangle\}$. Now, if we re-express the initial state using this basis of eigenstates for Particles 1 and 2, we find: $|\Psi\rangle_{123} = \tfrac{1}{2}\{|\Phi\rangle_+(\alpha|0_3\rangle - \beta|1_3\rangle) + |\Phi\rangle_-(\alpha|0_3\rangle + \beta|1_3\rangle) + |\Omega\rangle_+(-\alpha|1_3\rangle + \beta|0_3\rangle) + |\Omega\rangle_-(-\alpha|1_3\rangle - \beta|0_3\rangle\}$. Thus, after performing my measurement, I will "collapse" the system onto one of these four summands. Once I communicate to Nicholas (via ordinary means), which summand I find, he knows how to manipulate Photon 3 to reproduce the original state of Photon 1. For instance, if I find that my measurement yields state $|\Phi\rangle_-$, then Nicholas does not need to do anything to Photon 3, since, as above, it is already in the original state of Photon 1. If I find any other result, Nicholas will have to perform a